

Elmon Toraman

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Summary

Elmon is a geotechnical engineer based in Los Angeles. She specializes in multidisciplinary infrastructure projects, including railways, roadways, data centers, and energy developments. She brings expertise in geotechnical engineering, with experience spanning preliminary data reviews, desk studies, and field exploration management. She has a strong background in planning and interpreting site investigations, assessing soil and rock properties, and delivering design solutions for temporary works, shallow and deep foundations, retaining walls, and pile systems.

Elmon has extensive international experience, having worked in Türkiye, the UK, and the US. She has contributed to projects across the USA, Türkiye, Turkmenistan, Australia, Canada, Iraq, Saudi Arabia, Northern Cyprus, Pakistan, and Europe.

She is also experienced in coordinating multidisciplinary teams, ensuring quality control during fieldwork, and delivering technically sound geotechnical recommendations to support infrastructure and building projects. She has extensive experience leading geotechnical design for multiple data center projects and providing project management support for major data center clients.

Elmon applies advanced geotechnical modeling tools such as PLAXIS 2D and 3D to solve complex engineering challenges and predict ground movements related to excavation and tunneling activities. She is experienced in top-down construction design, integrating temporary works into permanent structures, coordinating contractors, preparing quantity take-offs, and bringing one year of full-time field engineering experience.

She also leads sustainability initiatives aimed at promoting sustainable practices within ground engineering projects, including carbon emission calculations and assessments to support low-carbon design strategies.

Education and Certifications

New York University, NY M.S. Applied Urban Science and Informatics <i>Merit Scholarship Recipient</i>	2020
University of California, Berkeley Visiting Researcher <i>Received Fulbright Scholarship to conduct research using machine learning models to assess the impact of soil conditions on earthquake-induced structural damage, enhancing accuracy in seismic hazard prediction.</i>	2012
Istanbul Technical University, Istanbul M.S. Geotechnical and Earthquake Engineering	2004
Yildiz Technical University, Istanbul B.S. in Civil Engineering	2001

Certifications

Professional Engineer (PE) Certification – Civil Geotechnical Discipline | Passed 8-hour exam

Professional Experience

Senior Geotechnical Engineer | Arup, Los Angeles, CA, USA

Oct 2021 – Nov 2025

- **Site Due Diligence for Confidential Data Center Projects Reno, NV; Tucson, AZ; San Antonio, TX; Dallas, TX**
Served as geotechnical lead for 15 potential data center sites. Elmon led the geotechnical exploration program and generation of desk study report and site due diligence reports. Studies covered a range of hazards, such as seismic activity, liquefaction potential, fault rupture, problematic soil, and landslides. Elmon also carried site coordinator role for these projects. In addition to due diligence desk studies, she led the team preparing geotechnical reports, providing design support and construction administration services.
- **ACS - Due Diligence Studies, Multiple Locations**
Acted as geotechnical lead for due diligence review of data-room contents for acquisition of up to 6 data center sites in the US.
- **Ontario Wind Diligence Studies, Ontario, Canada**
Geotechnical lead for due diligence works on the Nipigon Wind Development, a 200 MW wind energy facility located on Crown land in the District of Thunder Bay, Ontario.
- **Microsoft Cherry Valley FEP01 Site Due Diligence and Early Entitlements, Cherry Valley, IL**
Served as deputy project manager for multidisciplinary site due diligence for potential new construction of five 48 MW data centers on a more than 200-acre site. Supported entitlements team as Engineering Project Manager during Early Entitlements phase of project.
- **Quebec City Tramway OE Quebec City, Canada**
Provided technical support to the local team by developing geotechnical specifications, coordinating supplemental investigations, and interpreting lab results, enabling the identification of critical soil conditions and optimization of design.

solutions.

- **LA Exposition Park Subterranean, Los Angeles, CA**

Provided geotechnical support for the proposed \$300M, 3-level, 2,000-car subterranean parking structure at the southeast corner of Expo Park. Elmon's role included coordination of multiple subcontractors for site work including ground investigation and laboratory contractors, surveyors and environmental consultants. She coordinated laboratory testing programs, analyzed data, assessed soil parameters, performed foundation analysis, and provided excavation recommendations.

- **VTA BSVII CP2, San Francisco, CA**

Supported Property Protection team on impact assessment analysis including calibration of lab test results to the soil parameters for Plaxis HS model and numerical modeling of bridges. Developed 2D and 3D Plaxis analysis for bridges.

- **I-405 Improvement Project, Los Angeles, CA**

Geotechnical team member for design-build project for Orange County Transportation Authority (OCTA) to widen 16 miles of the I-405 in Orange County. She contributed to the design of bridge foundations, sound walls, and provided design and construction phase services, coordinating both the design and construction joint ventures, as well as Caltrans reviewers.

- **1200 15th St, San Francisco, CA**

Geotechnical Engineer for the concept-phase evaluation of a two-story facility founded on liquefiable soils, to be converted into office space for the San Francisco Municipal Transportation Agency. She developed soil improvement solutions and prepared specifications for measures such as compaction grouting to mitigate liquefaction.

- **LAX Receiving Station X (RS-X), Los Angeles International Airport (LAX), Los Angeles, CA**

Geotechnical Engineer for structures of the new 230/34.5kV Electrical Receiving Station (RS-X) for LADWP as a dedicated power supply facility for LAX.

- **LAWA Roadway, Utility, and Enabling (RUE) Projects, Los Angeles, CA**

Provided geotechnical support for light pole foundation design, roadway design for a series of Design-Build enabling projects that are critical to the successful delivery of the Los Angeles World Airports' (LAWA) Landside Access Modernization Program (LAMP).

Team Lead | Destech Consultancy, Istanbul, Turkiye

Oct 2018 – Sept2019

Managed time and budget for a diverse variety of projects including energy, infrastructure and transportation.

Associate, Geotechnical Engineer | Arup, Istanbul TR & London, UK

May 2005 –Sep 2018

Managed engineering team by assigning the workload, budgeting out time and costs.

Managed and coordinated field and laboratory activities, writing specifications and geotechnical design, technical report writing and supervision of project sites. Executed the design of projects ranging in construction value and complexity from \$5M to \$80M each. Supervised construction sites monitoring performance and compliance and ensuring workforce involvement.

Part Time Lecturer, Okan University, Istanbul, Turkiye

Jan 2019 – Aug 2020

Lectured "Soil Mechanics", "Foundation Engineering", and "In Situ Soil Testing" classes.

Product Manager | Geogrup Istanbul, Turkiye

May – Sept 2004

Provided technical assistance and sale of finite element softwares.

Selected Publications

- Fischer, E.; Hakhamaneshi, M.; Unal, M.; Alberto, Y.; Ay, D.; Chhabra, J.; Isufi, B.; Martin, A.; Nikolaou, S.; Rodriguez-Nikl, T.; Toraman, E.; Wu, K.; "Izmir Earthquake Phase 2 Report ", EERI VERT, 2020
- Toraman,E., Cebeci S., Gündüz A.; "Parametric Design of Piled Foundations and Automation ", ZMGM17, 2018
- Toraman,E., Ural D.; "Estimation of Structural Damage Due To Seismic Soil Parameters Following the Northridge Earthquake ", 16th World Conference on Earthquake Engineering, 2017
- Toraman,E., Ural D.; "Hazard estimation of Existing Buildings in the Cities of Kocaeli and Adapazari due to Local Soil Conditions ", IAHS, 2013